Trajectory Prediction for Maneuverable Target Based on Intention Recognition and Intercept Engagement

KAIZE YU
JIANGLONG YU
XIAODUO LI
XIWANG DONG, Senior Member, IEEE
Beihang University, Beijing, China

This article proposes a trajectory prediction algorithm based on target intention recognition and a distribution cropper for maneuverable targets. Different from the existing research, the trajectory prediction is decoupled into the structure of intention recognition and trajectory extrapolation to introduce intercept engagement and distribution cropper. First, the intention recognition and trajectory prediction are modeled to establish the structure. The intention recognition based on intercept engagement is utilized to enhance trajectory prediction accuracy. Second, the distribution cropper in the graph attention mechanism is established based on the pursuit–evasion decision. Unreasonable prediction results are filtered out through the interactive distribution cropper under the intercept engagement. Third, the trajectory prediction training process is supervised based on the proposed learning mechanism that incorporates intercept engagement and pursuit–evasion decision. This approach enhances the neural network training performance by incorporating the adversarial dynamics between the target and the defender aircraft. Finally, the effectiveness and feasibility of the proposed algorithm are demonstrated through numerical simulations under representative intercept engagement scenarios.

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Authors' addresses: Kaize Yu and Xiaoduo Li are with the Institute of Unmanned System, Beihang University, Beijing 100191, China, E-mail: (yukaize@buaa.edu.cn; lixiaoduo@buaa.edu.cn); Jianglong Yu is with the School of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China, E-mail: (sdjxyjl@buaa.edu.cn); Xiwang Dong is with the Institute of Unmanned System, School of Artificial Intelligence, and School of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China, E-mail: (xwdong@buaa.edu.cn). *(Corresponding author: Jianglong Yu.)*

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I. INTRODUCTION

In recent years, the development of highly maneuverable vehicles has advanced rapidly. The kinematic characteristics of this type of target exhibit nonlinearity, abrupt changes, and adversarial behavior during maneuvering [1], [2], resulting in significant uncertainty in trajectory prediction when relying only on kinematic modeling or single data-driven methods [3], [4]. Trajectory prediction plays a significant role in intercepting maneuverable targets, and related research is receiving widespread attention. There are primarily two paradigms for trajectory prediction. One approach is the trajectory extrapolation driven by physical prior information from a kinematic model. The other involves extracting spatio-temporal features of target trajectories using neural networks and establishing a trajectory prediction model through learning from a massive dataset. However, the existing approaches often result in significant trajectory prediction errors when dealing with noncooperative maneuverable target [5], [6]. The maneuvering and evasion strategies of the target are not generated independently, but in response to the intercept engagement through the pursuit–evasion decision process [7], [8], [9]. Therefore, it is particularly important to incorporate the intercept engagement as a feature into trajectory prediction while considering the pursuit–evasion decision process [10], [11], [12].

Trajectory prediction for maneuverable targets is subjected to strong interference coupling under time-varying kinematic trends. Existing methods often overlook decoupled trajectory analysis, leading to substantial prediction errors, and their performance is highly dependent on training data quality and scale, which limits practical applicability [13], [14], [15], [16]. By decoupling maneuver intention recognition from trajectory extrapolation, a modular prediction architecture can better capture maneuver semantics and markedly improve accuracy. Accordingly, establishing a decoupled trajectory prediction framework is both technically important and practically promising.

Maneuver intention recognition is a key component of the decoupled trajectory prediction framework. Prior work often formulates intention recognition as a clustering or sequence-learning problem and leverages recurrent neural network (RNN) and long short-term memory (LSTM) architectures to extract spatio-temporal features, learning noncooperative maneuver patterns from data [17], [18], [19]. To mitigate information deficiency, robust schemes have been proposed, including data processing and temporal modeling under limited observations, cross-scenario generalization with label normalization, and model inversion from observed trajectories [20], [21], [22], [23]. Nevertheless, most existing methods do not explicitly integrate

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intercept engagement factors or the target's pursuit–evasion decision process, leaving decision-oriented intention recognition under engagement constraints largely unexplored.

Trajectory extrapolation is another core component of the decoupled framework. Existing approaches fall into two categories. The first relies on filtering and kinematic modeling to fit and extrapolate motion trajectories, yet it often exhibits limited interpretability and degraded stability for maneuverable targets in adversarial settings [24], [25], [26], [27], [28]. The second formulates extrapolation as supervised learning and leverages neural predictors, where prior trajectories and kinematic states are commonly fed into RNN, LSTM, and gate recurrent unit (GRU) models, and performance can be further improved by multiscale convolutional–recurrent hybrids or constrained variants, such as constrained long short-term memory (cLSTM) [29], [30], [31], [32], [33], [34], [35], [36], [37]. However, for maneuverable noncooperative targets, trajectories can change abruptly due to strongly coupled factors, and methods driven solely by historical trajectories are often inadequate.

To mitigate this limitation, several studies augment the feature set with auxiliary cues that are not purely trajectory-derived. For instance, incorporating lane-change intention improves prediction in autonomous driving, but these findings are largely restricted to 2-D road scenarios [38], [39]. Using terminal positions as a feature in air traffic control (ATC) can increase candidate diversity, yet it typically assumes cooperative targets and fixed flight paths [40]. Maneuver categories or neighboring-aircraft kinematics have also been introduced as features, but existing formulations usually involve limited maneuver sets or remain tailored to cooperative, nonconfrontational settings [41], [42], [43]. Overall, trajectory prediction in the literature is still predominantly studied for cooperative targets or nonadversarial scenarios, and the complexity and uncertainty inherent to maneuverable noncooperative targets remain insufficiently addressed. Consequently, establishing a decoupled intention-recognition and trajectory-extrapolation architecture that explicitly incorporates intercept engagement and pursuit–evasion decision cues remain important and open.

For trajectory prediction of maneuverable noncooperative targets, the decoupled structure considering intercept engagement and pursuit–evasion decision is proposed in this article. Based on the existing research, the main contributions of this article are as follows. First, a decoupled trajectory prediction structure incorporating intention recognition and trajectory extrapolation is established to reduce interference coupling level and improve accuracy. In addition, the proposed independent intention recognition is based on intercept engagement and pursuit–evasion decision. Second, an interactive distribution cropper for trajectory prediction in graph attention network (GAT) is proposed to eliminate the unreasonable prediction results. Third, a supervised learning mechanism is developed to incorporate pursuit–evasion decision cues, thereby improving training effectiveness.

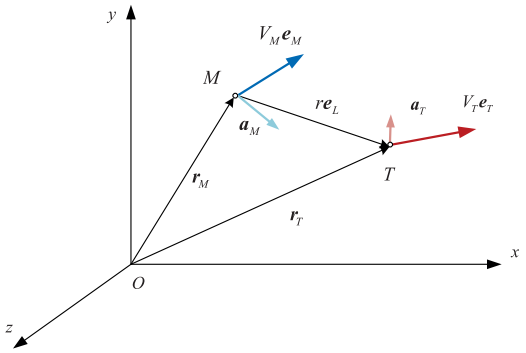


Fig. 1. 3-D guidance geometry of the target and the defender aircraft.

II. PROBLEM FORMULATION

Fig. 1 demonstrates the 3-D engagement geometry between the defender and the target. Let M and T denote the defender aircraft and the target, respectively. Let $\boldsymbol{r}_M$ and $\boldsymbol{r}_T$ denote the displacement vectors of the defender and the target, respectively. Let V_M and V_T denote the speeds, and let $\boldsymbol{e}_M$ and $\boldsymbol{e}_T$ denote the corresponding unit direction vectors of velocity. Let $\boldsymbol{a}_M$ and $\boldsymbol{a}_T$ denote the accelerations of the defender and the target, respectively. Nonlinear kinematic models for the defender aircraft and the target are established as follows:

$$\begin{aligned} \dot{\boldsymbol{r}}_M &= V_M \boldsymbol{e}_M, \dot{\boldsymbol{e}}_M = \boldsymbol{a}_M / V_M \\ \dot{\boldsymbol{r}}_T &= V_T \boldsymbol{e}_T, \dot{\boldsymbol{e}}_T = \boldsymbol{a}_T / V_T. \end{aligned} \tag{1}$$

To characterize the relative kinematics between the target and the defender, we derive the relative-motion equations as follows:

$$\begin{aligned} \dot{r} &= (V_T \boldsymbol{e}_T - V_M \boldsymbol{e}_M) \cdot \boldsymbol{e}_L \\ \dot{\boldsymbol{e}}_L &= [\boldsymbol{e}_L \times (V_T \boldsymbol{e}_T - V_M \boldsymbol{e}_M)] \times \boldsymbol{e}_L / r \\ V_L &= V_M \boldsymbol{e}_M \cdot \boldsymbol{e}_L - V_T \boldsymbol{e}_T \cdot \boldsymbol{e}_L \end{aligned} \tag{2}$$

where $\boldsymbol{r}$ is the relative separation, $\boldsymbol{e}_L$ is the line of sight (LOS) unit vector, and V_L is the relative speed along the LOS.

The maneuver intention is characterized using the target acceleration $\boldsymbol{a}_T$, whose radial and normal components, $\boldsymbol{a}_r$ and $\boldsymbol{a}_n$, are defined as

$$\boldsymbol{a}_r = (\boldsymbol{a}_T \cdot \boldsymbol{V}_T) / \boldsymbol{V}_T \tag{3}$$

$$\boldsymbol{a}_n = \|\boldsymbol{V}_T \times \boldsymbol{a}_T\| / \|\boldsymbol{V}_T\|. \tag{4}$$

The intention space is constructed by labeling maneuvers according to the target acceleration $\boldsymbol{a}_T$.

Trajectory prediction for a maneuverable target is formulated as learning a mapping from the existing $\boldsymbol{X}$ to the future trajectory $\boldsymbol{Y}$ as

$$\boldsymbol{Y} = f(\boldsymbol{X}) \tag{5}$$

where $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_{t_o}]$ denotes the observed trajectory sequence and $\boldsymbol{Y} = [\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_{t_p}]$ denotes the predicted future trajectory. Time points t_o and t_p denote the observation length and the prediction horizon, respectively.

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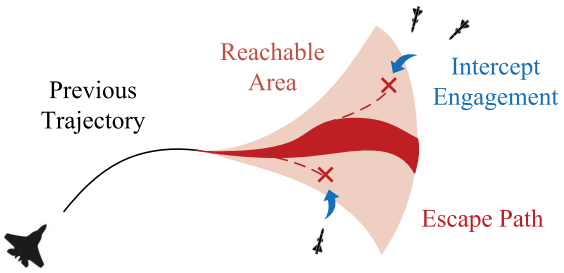


Fig. 2. Target escape reachable area.

This decoupled structure comprises two components: intention recognition and trajectory extrapolation. The trajectory prediction structure is established through decoupling analysis

$$\boldsymbol{I} = f_{\text{int}}(\boldsymbol{X}) \tag{6}$$

$$\boldsymbol{Y} = f_{\text{tra}}(\boldsymbol{X}, \boldsymbol{I}) \tag{7}$$

where the $\boldsymbol{I} = [I_1, I_2, \ldots, I_m]$ is the intention space of the target. The target maneuver acceleration $\boldsymbol{a}_T$ is labeled to establish the target intention space $\boldsymbol{I}$, where $I_k = f_I(a_n, a_r)$ is the maneuver intention. The mappings in (6) and (7) correspond to the intention-recognition module and the trajectory-extrapolation module, respectively. As shown in Fig. 2, trajectory prediction based solely on historical target kinematics may be insufficient in adversarial engagements. Accordingly, engagement-aware information is incorporated into the proposed formulation.

REMARK 1 For a noncooperative target, even with overload limits and terrain restrictions, the future trajectory can still diverge into a broad reachable set once the current motion state is given. When a pursuit–evasion decision is made in response to a specific intercept engagement, this reachable set is constrained to a narrower region, as illustrated in Fig. 2. The proposed framework incorporates engagement-driven pursuit–evasion decision-making together with maneuver intention recognition, thereby narrowing prediction hypotheses beyond what can be achieved using the observed trajectory alone and improving extrapolation accuracy. The intention space is defined at an abstract modeling level and can be instantiated either as discrete semantic maneuver classes or as ordered maneuver-intensity levels obtained by quantizing continuous acceleration, enabling the same formulation to infer both maneuver type and maneuver intensity.

By incorporating intercept engagement and pursuit–evasion decision, trajectory prediction accuracy is enhanced at the game theory and decision-making levels. The prediction accuracy is enhanced in the decoupled trajectory prediction by optimizing the factors Q_{int} and Q_{tra}, which are related to the target–defender dynamics

$$Q_{\text{int}} = \Theta_{\text{int}}(r, \dot{r}, \boldsymbol{e}_L, \dot{\boldsymbol{e}}_L) \tag{8}$$

$$Q_{\text{tra}} = \Theta_{\text{tra}}(r, \dot{r}, \boldsymbol{e}_L, \dot{\boldsymbol{e}}_L). \tag{9}$$

By introducing intercept engagement and pursuit–evasion decision, the problem in (6) and (7) can be transformed into

$$\hat{\boldsymbol{I}} = f_{\text{int}}(\boldsymbol{X}, Q^*_{\text{int}}) \tag{10}$$

$$\hat{\boldsymbol{Y}} = f_{\text{tra}}(\boldsymbol{X}, \hat{\boldsymbol{I}}, Q^*_{\text{tra}}). \tag{11}$$

From the perspective of the target, the relationship between intercept engagement kinematic state $\boldsymbol{s}_L = [\boldsymbol{r}, V_L, \boldsymbol{e}_L]$ and the intercept engagement factors Q_{int} and Q_{tra} can be established through neural networks.

REMARK 2 The intercept engagement factors Q_{int} and Q_{tra} are defined as compact representations characterized by the engagement kinematic state $\boldsymbol{s}_L = [\boldsymbol{r}, V_L, \boldsymbol{e}_L]$ and its temporal variations, describing the target–defender relative geometry and its evolution. Incorporating engagement factors into intention recognition and trajectory prediction reduces ambiguity in identifying similar maneuver intentions and suppresses unreasonable predictions. In practice, the relative information required for $\boldsymbol{s}_L$ is estimated from sensor measurements via trajectory tracking, and the corresponding temporal variations are obtained accordingly, completing the computational chain from observations to Q_{int} and Q_{tra}.

The objective of this article is to develop an engagement-aware trajectory-prediction framework for maneuverable targets in adversarial air engagements. Specifically, this article aims to 1) achieve independent intention recognition under the decoupled structure in (10) and (11) and integrate its outputs into trajectory extrapolation; 2) formalize and incorporate engagement-related information into intention recognition in (10) and trajectory extrapolation in (11), where Q_{int} and Q_{tra} serve as a compact notation for the corresponding engagement-related factors; 3) optimize Q_{int} and Q_{tra} to obtain Q^*_{int} and Q^*_{tra} with sufficient consideration of intercept engagement and pursuit–evasion decisions.

III. MAIN RESULTS

The structure of the proposed algorithm is illustrated in Fig. 3. The algorithm input drives intention recognition and the interactive distribution cropper. Intention recognition outputs an intention posterior that conditions cropping and extrapolation. The cropper generates candidates, builds an interaction graph, and applies graph-attention message passing to support cropping and blocking of engagement-inconsistent branches, producing engagement-consistent candidates. The trajectory extrapolator combines the cropped candidates with the intention posterior to generate the predicted trajectory. The supervised learning mechanism is a training-time pathway that shapes attention and cropping via teacher score construction, factor-specific supervision assignment, and loss stabilization. Online iterative update closes the loop by re-estimating intention and reapplying cropping before each forecast.

Fig. 4 summarizes the decoupled algorithm, consisting of an intention-recognition stage and a trajectory-extrapolation stage, where the latter integrates interactive

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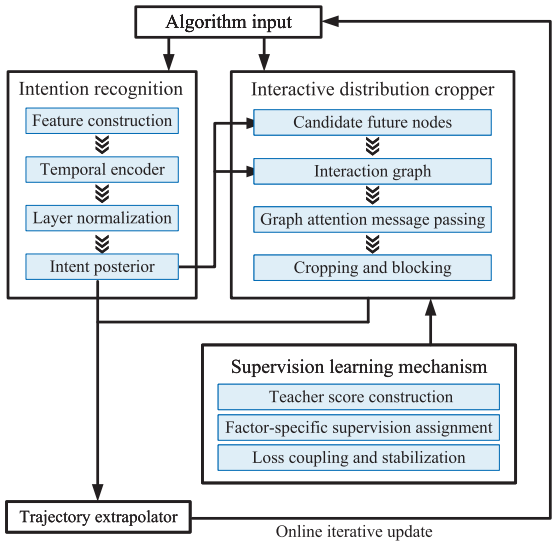


Fig. 3. Structured presentation of trajectory prediction algorithm.

distribution cropping and a training-time supervision mechanism for shaping attention behaviors. In a closed-loop interception-evasion interaction, the target maneuver cannot be extrapolated reliably from its past trajectory alone, because the evolving engagement geometry and pursuit–evasion decisions can rapidly redirect the future motion. As a result, even under physical constraints, the reachable future set remains highly branched. The proposed method is designed to reduce this branching ambiguity by incorporating engagement-driven interaction information through intention inference and engagement-consistent hypothesis pruning, and by updating predictions online at a short interval to remain synchronized with the fast-varying confrontation state.

A. Intention Recognition With BiLSTM Structure Based on Intercept Engagement

The intention recognition-based solely on prior information derived from trajectory observations struggles to distinguish similar intention with identical preparatory trajectory. To address this issue, the kinematic state of the target–defender dynamics is incorporated into the intention recognition. The range of possible target trajectories can be effectively constrained under specific intercept engagement. The intention recognition in (10) is transformed into a clustering problem and processed using bidirectional long short-term memory (BiLSTM) structure. The BiLSTM-based intention recognition can effectively handle situations involving sudden target maneuvers or abrupt trajectory changes.

The kinematic state $\boldsymbol{s}_M = [\boldsymbol{r}_M, V_M, \boldsymbol{e}_M]$ of the defender aircraft is used as the intention recognition factor Q_{int} and incorporated as part of the feature extraction process. The target trajectory sequence $\boldsymbol{X}$ and defender aircraft kinematic state $\boldsymbol{s}_M$ undergo relativization and normalization processing to yield $\boldsymbol{x}$ and $\tilde{\boldsymbol{s}}_M$. The trajectory sequences $\boldsymbol{x}$ and defender aircraft kinematic states $\tilde{\boldsymbol{s}}_M$ are processed using BiLSTM and fully connected layers

$$\boldsymbol{h}_t = \text{ReLU}(\boldsymbol{W}_t \text{BiLSTM}(\boldsymbol{x}) + \boldsymbol{b}_t) \tag{12}$$

$$\boldsymbol{h}_s = \text{ReLU}(\boldsymbol{W}_s \tilde{\boldsymbol{s}}_M + \boldsymbol{b}_s) \tag{13}$$

where $\boldsymbol{h}_t$ denotes the concatenated hidden state and $\boldsymbol{h}_s$ represents the result of static state encoding of $\tilde{\boldsymbol{s}}_M$ by the fully connected layer. Rectified linear unit (ReLU) is employed to endow the network with nonlinearity and sparse activations while maintaining stable gradients for efficient training. $\boldsymbol{W}_t$ and $\boldsymbol{b}_t$ represent the coefficients and biases for the sequence branch fully connected layer, respectively. $\boldsymbol{W}_s$ and $\boldsymbol{b}_s$ denote the coefficients and biases for the static branch, respectively.

After the splicing process in (12) and (13), a Softmax layer outputs the intention probability distribution

$$\hat{\boldsymbol{p}} = \text{Softmax}\left[\boldsymbol{W}_c \text{ReLU}(\boldsymbol{W}_h \boldsymbol{z} + \boldsymbol{b}_h) + \boldsymbol{b}_c\right] \tag{14}$$

where $\hat{\boldsymbol{p}} = [\hat{p_1}, \hat{p_2}, \ldots, \hat{p_m}]$ is the probability distribution of intention and $\hat{p_k}$ corresponds to the probability of I_k in the intention space $\boldsymbol{I}$. The $\boldsymbol{z} = [\boldsymbol{h}_t; \boldsymbol{h}_s]$, $\boldsymbol{W}_h$ and $\boldsymbol{b}_h$ represent the first layer of the classification head, while $\boldsymbol{W}_c$ and $\boldsymbol{b}_c$ represent the second layer. The $\hat{\boldsymbol{p}}$ denotes the probability distribution of the intention label.

B. Trajectory Interactive Distribution Cropper of GAT Based on Target–Defender Dynamics

Intercept engagement and pursuit–evasion decisions strongly shape target maneuvers and the resulting trajectories. The GAT is introduced within the interactive cropper module to model candidate future nodes and their relational structure. Attention-based message passing over the candidate graph encodes engagement-induced correlations into adaptive aggregation weights, thereby enabling engagement-consistent node selection and improving trajectory extrapolation.

The structure of the GAT is shown in Fig. 5. The attention mechanism $\alpha_{ij}^{(\ell)}(\boldsymbol{W}_\ell \boldsymbol{h}_i^{(\ell)}, \boldsymbol{W}_\ell \boldsymbol{h}_j^{(\ell)})$ is employed and parameterized by a weight vector $\boldsymbol{a}_\ell$. The multihead attention with $K = 3$ heads and node 1 is used as an example. The aggregated features from each attention head are concatenated or averaged to obtain $\boldsymbol{h}_i^{(\ell+1)}$. $\boldsymbol{h}_i^{(\ell)}$ denotes the node embedding of node i at GAT layer ℓ, and the node indices, as shown in Fig. 5, illustrate a single example neighborhood of the central node rather than the total graph size.

The response of the target to the intercept engagement can be modeled using the interactive graph. Information is exchanged between adjacent nodes through graph edges. After obtaining the intention recognition results, the target–defender confrontation interactive graph is established as

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \varphi_{\text{edg}}) \tag{15}$$

where $\mathcal{V}$ is the node set describing the engagement within one observation window. Specifically, $\mathcal{V}$ contains a set of target-trajectory nodes that encode the temporally ordered target states in the observation window, and a set of defender-state nodes that encode the corresponding defender kinematic states used to represent target–defender

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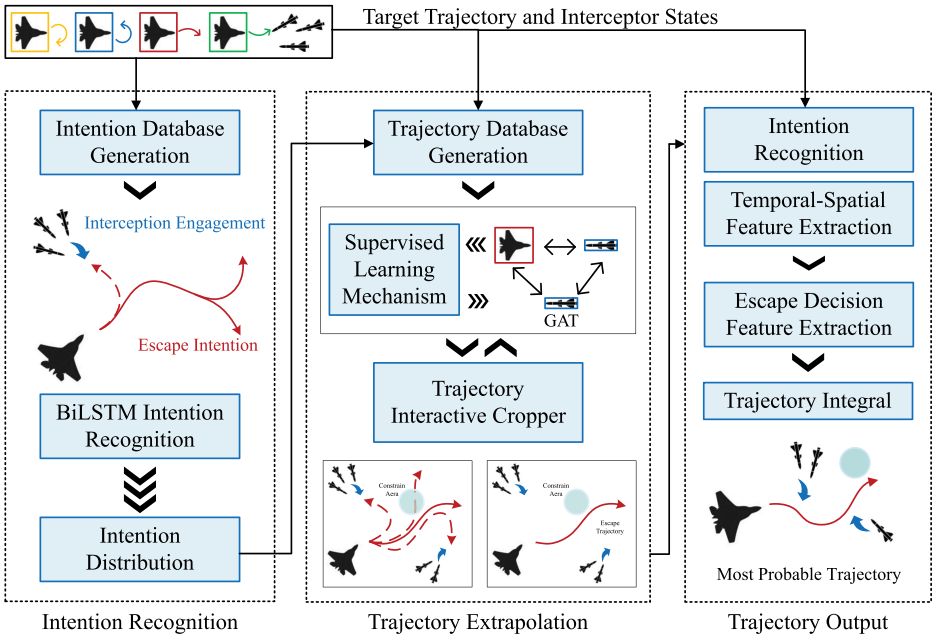


Fig. 4. Decoupled intention recognition and trajectory extrapolation structure.

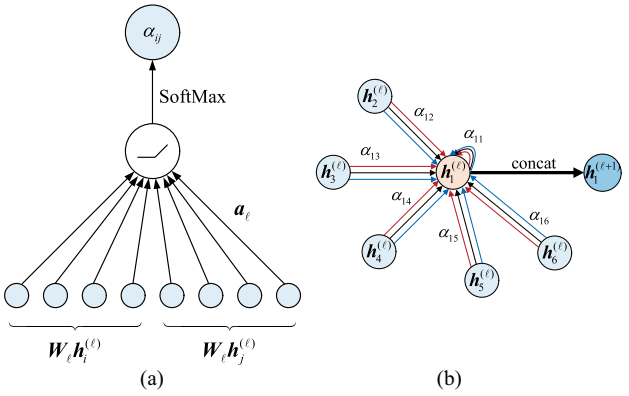


Fig. 5. Structure of GAT. (a) Single-layer attention network. (b) Multihead attention.

confrontation. Let $t_o = T_o/\Delta t$ denote the number of sampled steps in the observation window. Then, the interaction graph contains t_o target nodes and t_o defender nodes, and the total number of nodes is $|\mathcal{V}| = 2t_o$ for each update instant.

The edge set $\mathcal{E}$ specifies how information is allowed to propagate among nodes and is composed of three edge types. Temporal edges connect consecutive nodes along the same entity timeline, enforcing chronological consistency of either the target sequence or the defender sequence across the observation window. Spatial edges connect target nodes and defender nodes at the same time index, encoding instantaneous target–defender coupling induced by the engagement geometry. Countermeasure edges connect nodes to indicate interaction constraints imposed by intercept engagement and the evasion mechanism, so that feasibility-related information can modulate attention propagation in the subsequent cropping stage.

The edge feature mapping φ_{edg} assigns each edge an engagement descriptor composed of relative separation, LOS-relative speed and its temporal variation, LOS direction and its temporal variation, and a visibility indicator. The visibility indicator is determined by the target field-of-view constraint and is used to identify whether the defender is observable from the target at the current geometry.

Hidden states of the observed target trajectory $\boldsymbol{h}_t$, the defender kinematic state $\boldsymbol{h}_s$, and the intention-recognition output $\hat{\boldsymbol{p}}$ are fused as follows:

$$\boldsymbol{h}^{(0)}(k) = \Phi\left(\boldsymbol{g}^{(0)}(k), \left[\boldsymbol{h}_t; \boldsymbol{h}_s; \hat{\boldsymbol{p}}\right]\right). \tag{16}$$

When propagating features across the layer ℓ of the GAT, each feature is updated with weights based on information from neighboring nodes

$$\hat{\boldsymbol{h}}_i^{(\ell+1)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij} \boldsymbol{W} \boldsymbol{h}_j^{(\ell)}\right) \tag{17}$$

$$\boldsymbol{h}_i^{(\ell+1)} = \mathrm{LN}\left(\hat{\boldsymbol{h}}_i^{(\ell+1)} + \boldsymbol{h}_i^{(\ell)}\right) \tag{18}$$

where $\hat{\boldsymbol{h}}_i^{(\ell)}$ is the hidden state of node i in layer ℓ, α_{ij} is the attention weight, $\sigma(\cdot)$ is the nonlinear activation function, $\mathcal{N}(i)$ is the set of neighbors of node i, and LN performs normalization to ensure training stability. To ensure consistency in physical calculations and environments, soft masking calculations are performed prior to attention normalization

$$m_{ij}^{(\ell)} = \sigma\left(f_e^{(\ell)}\left(\boldsymbol{h}_i^{(\ell)}, \boldsymbol{h}_j^{(\ell)}, \boldsymbol{u}\right)\right) \tag{19}$$

$$m_j^{(\ell)} = \sigma\left(f_n^{(\ell)}\left(\boldsymbol{h}_j^{(\ell)}, \boldsymbol{u}\right)\right) \tag{20}$$

where $\boldsymbol{u}$ denotes an engagement descriptor associated with the LOS-related state $\boldsymbol{s}$, which compactly summarizes LOS

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geometry and visibility constraints relevant to interaction modeling. The functions $f_e^{(\ell)}(\cdot)$ and $f_n^{(\ell)}(\cdot)$ are learnable evaluation mappings that take the node hidden states together with $\boldsymbol{u}$ as inputs and output scalar gating scores for edgewise and nodewise soft masking, respectively. Specifically, $f_e^{(\ell)}$ produces the edge mask $m_{ij}^{(\ell)}$ based on the interaction between nodes i and j, while $f_n^{(\ell)}$ produces the node mask $m_j^{(\ell)}$ for node-level conditioning; both masks constrain message passing prior to attention normalization. This yields the GAT weight, which is calculated as follows:

$$\alpha_{ij}^{(\ell)} = \frac{m_{ij}^{(\ell)} m_j^{(\ell)} \exp(\mathrm{LR}(\boldsymbol{\alpha}_\ell^{\mathrm{T}}[\boldsymbol{W}_\ell \boldsymbol{h}_i^{(\ell)} \| \boldsymbol{W}_\ell \boldsymbol{h}_j^{(\ell)}]))}{\sum_{k\in\mathcal{N}(i)} m_{ik}^{(\ell)} m_k^{(\ell)} \exp(\mathrm{LR}(\boldsymbol{\alpha}_\ell^{\mathrm{T}}[\boldsymbol{W}_\ell \boldsymbol{h}_i^{(\ell)} \| \boldsymbol{W}_\ell \boldsymbol{h}_k^{(\ell)}]))} \tag{21}$$

where the $\boldsymbol{W}_\ell$ is the feature transformation matrix and $\boldsymbol{\alpha}_\ell$ is the attention parameter vector. The LR is the LeakyReLU layer. This mechanism allows for the full extraction of temporal and spatial features inherent to the input sequence, consistency with neighboring points, and interaction constraints with defender aircraft during training.

The impact of intercept engagement on target trajectory is represented by the GAT interactive distribution cropper. The distribution cropper is applied during attention propagation in the GAT to minimize trajectory extrapolation divergence. The cropping of unreasonable trajectory is achieved through the redistribution of attention weight. The interactive distribution cropper constraints are defined from aspects related to confrontation with target and defender aircraft as follows:

$$\mathcal{C}_{\mathrm{esc}} = \left\{ \hat{\boldsymbol{y}}_j^{(i)} | \dot{r} \geq \dot{r}_{\min} \right\} \tag{22}$$

$$\mathcal{C}_{\mathrm{env}} = \left\{ \hat{\boldsymbol{y}}_j^{(i)} \mid g(\hat{\boldsymbol{y}}_j^{(i)}) \notin \mathcal{F} \right\} \tag{23}$$

$$\mathcal{C}_{\mathrm{ene}} = \left\{ \hat{\boldsymbol{y}}_j^{(i)} \mid \frac{1}{T} \sum_{t_0}^{t_0+T} \left\| \boldsymbol{a}_{T,k} - \boldsymbol{a}_{T,k-1} \right\| \leq a_{T,\max} \right\} \tag{24}$$

where $\hat{\boldsymbol{y}}_j^{(i)}$ represents the trajectory of the sample i at node j, $r_{\min}$ denotes the distance constraint, and $a_{T,\max}$ is the acceleration constraint. $g(\cdot)$ maps a three-dimension position to the two-dimension grid-cell index on the ground map. $\mathcal{F} \subset \mathbb{Z}^2$ is the set of free grid cells satisfying terrain, obstacle, no-fly, and altitude-clearance requirements. The condition $g(\hat{\boldsymbol{y}}_j^{(i)}) \notin \mathcal{F}$ indicates that the trajectory violates the terrain constraint at step k, hence $\hat{\boldsymbol{y}}_j^{(i)} \in \mathcal{C}_{\mathrm{env}}$. In the overall algorithm, $\mathcal{C}_{\mathrm{env}}$ is used to filter out infeasible samples or impose additional penalties during trajectory selection. If the target trajectory $\hat{\boldsymbol{y}}_j^{(i)}$ after maneuvering exhibits $\dot{r} < 0$ and a continuously increasing trend, then it is classified as a nonevasive maneuver and removed from the result. The $a_{T,\max}$ is used to restrict energy dissipation during maneuvres. The set of interactive distribution cropper constraints is $\mathcal{C} = \mathcal{C}_{\mathrm{esc}} \cup \mathcal{C}_{\mathrm{env}} \cup \mathcal{C}_{\mathrm{ene}}$ and the attention weight in (21) is redistributed as

$$\alpha_{ij}^{(\ell)} \leftarrow \alpha_{ij}^{(\ell)} \cdot \mathbf{1}_c(j) \tag{25}$$

where the indicator function $\mathbf{1}_c(j)$ indicates whether node j satisfies the constraint, serving as a feasibility indicator for the attention mask. The above process suppresses unreasonable nodes and edges, and prevents the propagation of erroneous information to neighboring nodes. The distribution cropper algorithm is shown in Table I. This allows trajectory extrapolation to be based on historical data while fully considering the effects of the intercept

TABLE I
Distribution Cropper Based on Pursuit–Evasion Decision and Intercept Engagement

Algorithm 1

Input: $\mathcal{N}(i)$, $\dot{r}_{\min}$, $\mathcal{F}$, $a_{T,\max}$, and T.
Output: Cropped edge weights $\alpha_{ij}^{(\ell)}$ and feasibility indicators $m(j)$.

Step 1: Input: The neighbor set $\mathcal{N}(i)$ of node i, the escape constraint threshold $\dot{r}_{\min}$, the unreachable set $\mathcal{F}$ and the acceleration threshold $a_{T,\max}$. The candidate target trajectory kinematic state are selected from neighbor j to node i at layer ℓ with the sampling window T.

Step 2: The initial compatibility score $e_{ij}^{(\ell)}$ is calculated based on the $\boldsymbol{q}_i^{(\ell)}$, $\boldsymbol{k}_j^{(\ell)}$ and $\boldsymbol{v}_j^{(\ell)}$

$$e_{ij}^{(\ell)} = \left(\boldsymbol{q}_i^{(\ell)}\right)^{\mathrm{T}} \boldsymbol{k}_j^{(\ell)} / \sqrt{d_k} + b\left(\boldsymbol{r}_{ij}\right). \tag{26}$$

The initial edge weights $\alpha_{ij}^{(\ell)}$ are output through the Softmax layer according to (21).

Step 3: The constraint check is performed after applying the differential operation to the candidate trajectory $\hat{\boldsymbol{y}}_j^{(i)}$.
1) Pursuit-evasion decision and evasion tendency determination based on the constraints in (22)

$$c_{\mathrm{esc}}(j) = 1 \cdot [\frac{1}{T} \sum_k \dot{r}_k \geq \dot{r}_{\min}]. \tag{27}$$

2) Determine environmental and terrain constraints based on the constraints in (23)

$$c_{\mathrm{env}}(j) = 1 \cdot [g(\hat{\boldsymbol{y}}_j^{(i)}) \notin \mathcal{F}, \forall k]. \tag{28}$$

3) Determine energy loss and maneuvering constraints based on the constraints in (24)

$$c_{\mathrm{ene}}(j) = 1 \cdot [\frac{1}{T} \sum_{k=t_0}^{t_0+T} \left\| \boldsymbol{a}_{T,k} - \boldsymbol{a}_{T,k-1} \right\| \leq a_{T,\max}]. \tag{29}$$

The feasibility indicator for neighbor j is synthesized based on the determination result

$$m(j) = c_{\mathrm{esc}}(j) \cdot c_{\mathrm{env}}(j) \cdot c_{\mathrm{ene}}(j) \in \{0, 1\}. \tag{30}$$

Step 4: The visibility indicator FOV is encoded in (15). If the defender aircraft is not visible according to the FOV indicator, the cropping is performed using only the engagement-independent constraint subset, with the environmental and terrain constraint in (23).

Step 5: The GAT edge weights are normalized according to (25) after interactive pruning based on feasibility indicators

$$\alpha_{ij}^{(\ell)} \leftarrow \alpha_{ij}^{(\ell)} m_j \exp\left(-\lambda_{\mathrm{crop}} \rho_{ij}\right) / S_i, \tag{31}$$

where the $S_i = \sum_{k\in\mathcal{N}(i)} \alpha_{ik}^{(\ell)}$, the ρ_{ij} is the soft penalty and the λ_{crop} is the crop intensity.

Step 6: The hidden layer state $\boldsymbol{h}_i^{(\ell)}$ and the corresponding $\boldsymbol{z}_i^{(\ell)}$ are updated based on information aggregation.

Step 7: The loss is trained based on $\boldsymbol{h}_i^{(\ell)}$ and $\boldsymbol{z}_i^{(\ell)}$ according to supervised learning.

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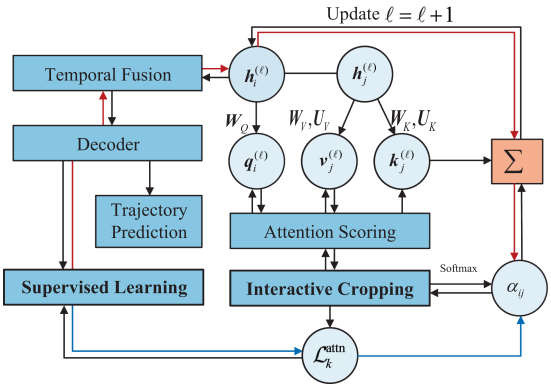


Fig. 6. GAT with interactive distribution cropper and supervised learning.

engagement. The update function for the hidden layer $\boldsymbol{h}_i$ in (17) changes accordingly. The specific implementation process is shown in Fig. 6. The interactive cropper follows the attention scoring. The edge weight α_{ij} is modified based on multiple factors related to pursuit–evasion decision and intercept engagement. The unreasonable trajectories are ultimately filtered through edge weights α_{ij}. The supervised learning mechanism encompasses the attention layer and the geometric layer. After the extrapolation results are obtained through the decoder, the loss function of the attention mechanism flows back through the red and blue pathways. The loss function is defined in (54) and (55). The red and blue paths converge after reaching α_{ij}, jointly influencing the $\boldsymbol{W}$ and $\boldsymbol{U}$ used to compute $\boldsymbol{q}_i^{(\ell)}$, $\boldsymbol{v}_j^{(\ell)}$, and $\boldsymbol{k}_j^{(\ell)}$. The optimization of GAT is achieved through changes in the gradient.

The features of each extrapolation node are mapped to future position estimates after internal propagation through multilayer graph attention. The trajectory incremental sequence achieves spatio-temporal fusion through causal self-attention and is output by the decoder

$$\Delta\hat{\boldsymbol{y}}_k = \mathrm{MLP}\left(\boldsymbol{h}_k^L\right) \tag{32}$$

where the prediction time $k = 1, 2, \ldots, t_{\mathrm{p}}$ and the MLP is the multilayer perceptron used for spatio-temporal fusion and decoding. The generated sequence of coordinate increment sequence $\{\Delta\hat{\boldsymbol{y}}_k\}$ is integrated starting from the initial position $\boldsymbol{r}_{T,t_o}$ to obtain the extrapolation result

$$\hat{\boldsymbol{y}}_t = \sum_{k=0}^{t} \Delta\hat{\boldsymbol{y}}_k + \boldsymbol{r}_{T,t_o}. \tag{33}$$

The output trajectory is $\hat{\boldsymbol{Y}} = \{\hat{\boldsymbol{y}}_t\}$, where $t = 1, 2, \ldots, t_p$. In the trajectory extrapolation, the introduction of intention recognition and intercept engagement is achieved through GAT and distribution cropper. The trajectory extrapolation-based solely on the observed trajectory leads to results that align with historical data but fail to reflect the current confrontation engagement. In this article, the intercept engagement that reflects confrontation is extracted as a feature and the trajectory distribution is cropped based on pursuit–evasion decision.

REMARK 3 The interactive cropper represents engagement evolution by an interaction graph so that target–defender dynamics are encoded as relations rather than being treated as independent feature channels. In this formulation, nodes correspond to candidate future states, and edges encode pairwise interaction cues induced by the engagement geometry. The GAT is adopted to perform attention-based message passing, where each node aggregates information from its neighbors through data-dependent attention weights. This mechanism enables adaptive weighting of interaction-relevant information and preserves permutation invariance with respect to the ordering of candidates, which is well suited to graph-structured candidate sets. Compared with fixed-weight aggregation or sequence-only models, GAT provides an explicit and interpretable way to capture relational correlations among candidate nodes, thereby supporting engagement-consistent node selection and improving the robustness of trajectory extrapolation under evolving confrontation geometry.

C. Supervised Learning Mechanism Based on Pursuit–Evasion Decision Against Intercept Engagement

The supervised learning mechanism is established based on the target pursuit–evasion decision under intercept engagement. The average displacement error (ADE) and final displacement error (FDE) metrics are typically used as supervisory indicators when handling this type of supervised learning

$$\mathrm{ADE} = \frac{1}{t_p}\sum_{t=1}^{t_p} \left\|\hat{\boldsymbol{y}}_t - \boldsymbol{y}_t\right\|_2 \tag{34}$$

$$\mathrm{FDE} = \left\|\hat{\boldsymbol{y}}_{t_p} - \boldsymbol{y}_{t_p}\right\|_2. \tag{35}$$

The supervised learning-based solely on the ADE and FDE neglects the target–defender confrontation and the effect of the pursuit–evasion decision. Ultimately, the training mechanism is unable to correlate intercept engagement with the dynamic response of the target. The advantage of employing GAT is that supervised learning can be established at the attention level for specific intercept engagement.

To enable factor-specific supervision and align the teacher scoring signals with the attention mechanism, the attention heads are partitioned into mutually exclusive groups.

DEFINITION 1 In the proposed supervised learning mechanism, the task division among attention heads is not a post hoc observation and does not occur by chance. Instead, it is explicitly specified by defining a fixed partition of the attention-head set $\mathcal{H}$ into three disjoint groups

$$\mathcal{H} = \mathcal{H}_E \cup \mathcal{H}_R \cup \mathcal{H}_T$$

where $\mathcal{H}_E \cap \mathcal{H}_R = \mathcal{H}_E \cap \mathcal{H}_T = \mathcal{H}_R \cap \mathcal{H}_T = \emptyset$. This partition is an engineering design that assigns different training objectives to different head groups. The group $\mathcal{H}_E$ is designated to learn attention behaviors guided by the teacher

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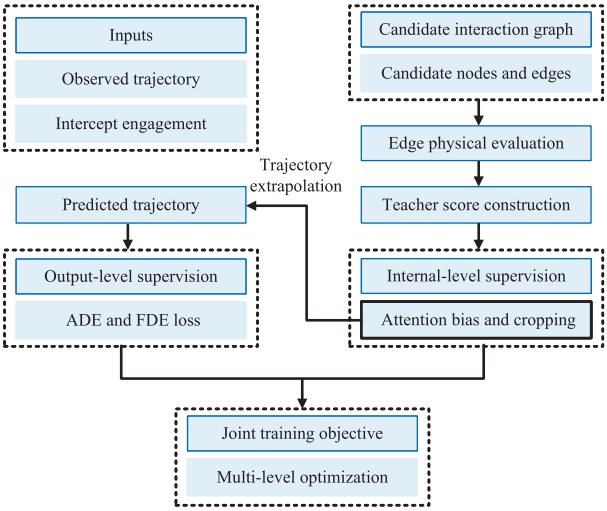


Fig. 7. Hierarchy of the supervised learning mechanism.

score for energy dissipation, the group $\mathcal{H}_R$ is designated to learn attention behaviors guided by the teacher score for interception risk, and the group $\mathcal{H}_T$ is designated to learn attention behaviors guided by the teacher score for task timeliness. During training, supervision and normalization are applied to the attention weights produced by each group using its corresponding teacher score and its corresponding edge subset. This factor-specific supervision enforces the intended functional roles of the head groups and prevents the attention-head division from degenerating into an incidental correlation.

Teacher scoring is introduced to provide supervision for attention allocation and node-retention decisions in the interactive cropper. Teacher scores are defined as constructed supervisory signals derived from engagement-related information, and are used to provide factor-specific guidance for learning attention under physical and mission constraints. They are necessary because trajectory-level distance losses do not explicitly encode feasibility requirements and may permit physically implausible attention allocation and candidate selection under adversarial engagements. Functionally, the teacher scores assign a scalar desirability score to each candidate transition, so that attention learning and interactive cropping are guided toward engagement-consistent candidates by emphasizing energy-efficient and smooth maneuvers, suppressing high-risk interception branches, and enforcing timeliness requirements.

The hierarchy in Fig. 7 clarifies how the proposed supervised learning mechanism operates beyond trajectory-level distance fitting. Output-level supervision is provided by ADE and FDE, which penalizes geometric deviations between the predicted trajectory and the ground truth. Internal-level supervision is provided by teacher scores constructed from engagement-related physical factors and applied to attention bias and interactive cropping, so that candidate transitions inconsistent with the current confrontation state are suppressed before trajectory extrapolation. By jointly optimizing these two supervision paths, the training process not only improves endpoint accuracy but also enforces engagement-consistent selection and message passing, thereby coupling intercept engagement with the target's dynamic response.

Various pursuit–evasion decision factors are projected onto the attention space to establish a teacher distribution model. The mutual influence between attention head groups is restrained by cross-group separation constraints. The obtained teacher distribution model is as follows:

$$\pi_{ij}^{(k),t} = \frac{\exp\left(s_{ij}^{(k),t}/\tau_k\right)\mathbf{1}\left[i \to j \in \mathcal{E}_k(i)\right]}{\sum_{k' \in \mathcal{E}_k(i)} \exp\left(s_{ik'}^{(k),t}/\tau_k\right)} \tag{36}$$

where $s_{ij}^{(k),t}$ is the score of edge $\mathcal{E}_{ij}$ and τ is the temperature factor. The intragroup attention loss function for each category k is formulated as

$$\mathcal{L}_k^{\text{attn}} = \sum_t \left\{ \frac{1}{|\mathcal{H}_k|} \sum_{h_k \in \mathcal{H}_k} D_{\text{KL}}\left(\pi_{ij}^{(k),t} \tilde{\beta}_k^{h_k,t}\right) + \rho_k \Delta H_k + \zeta_k \Delta \bar{\beta}_k \right\} \tag{37}$$

where

$$\Delta H_k = |H_{\min}^{(k)} - H\left(\bar{\beta}^{t,(k)}\right)| \tag{38}$$

$$\Delta \bar{\beta}_k = \left\| \bar{\beta}^{t,(k)} - \bar{\beta}^{t-1,(k)} \right\|_1 \tag{39}$$

which is used to prevent attention collapse within groups and the third term suppresses temporal jitter, respectively. The $\tilde{\beta}_k^{h_k,t}$ denotes the mask-normalized attention of the head h_k in the group $\mathcal{H}_k$ at time t, the $|\mathcal{H}_k|$ is the number of attention heads in the group $\mathcal{H}_k$, D_{KL} denotes the Kullback-Leibler KL divergence, H$(\cdot)$ denotes the corresponding entropy, $H_{\min}^{(k)}$ is the lower bound of the entropy for the group $\mathcal{H}_k$, and $\rho_k, \zeta_k \geq 0$ are the weights for controlling anticollapse and time smoothing, respectively. The $\bar{\beta}^{t,(k)}$ is calculated as

$$\bar{\beta}^{t,(k)} = \sum_{h_k \in \mathcal{H}_k} \tilde{\beta}_k^{h_k,t} / |\mathcal{H}_k| . \tag{40}$$

In addition to $\mathcal{L}_k^{\text{attn}}$, the loss function of the attention layer also include $\mathcal{L}_{\text{sep}}$ to prevent information coupling between groups. The information decoupling achieved through cross-group separation regularization ensures that attention heads do not interfere with each other

$$\mathcal{L}_{\text{sep}} = \eta \sum_k \sum_{h_k \in \mathcal{H}_k} \sum_{h_k' \notin \mathcal{H}_k} \cos^2\left(\text{vec}(\tilde{\beta}^{h_k}), \text{vec}\left(\tilde{\beta}^{h_k'}\right)\right) \tag{41}$$

where $\tilde{\beta}^{h_k}$ represents the attention tensor of the attention head h_k on the full temporal correlation edge and the function vec$(\cdot)$ is used to flatten the tensor into a vector in a fixed order. The use of the cross-group separation regularization mechanism effectively prevents mutual interference between attention groups during training.

1) *Energy loss and overload limit scoring:* The incremental sequence of the target position obtained from the

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trajectory extrapolation result in (32) can be third differentiated with respect to time

$$\boldsymbol{V}_{T,t} = \frac{\Delta \boldsymbol{y}_t}{\Delta t} \tag{42}$$

$$\boldsymbol{a}_{T,t} = \frac{\boldsymbol{V}_{T,t} - \boldsymbol{V}_{T,t-1}}{\Delta t} \tag{43}$$

$$\boldsymbol{j}_t = \frac{\boldsymbol{a}_{T,t} - \boldsymbol{a}_{T,t-1}}{\Delta t} \tag{44}$$

where the $\boldsymbol{j}_t$ is the jerk obtained by differentiating the target acceleration with respect to time at time t. The restriction on $\boldsymbol{j}_t$ during neural network training enables smoother output trajectories.

In addition to trajectory-level distance errors, the training objective incorporates supervision terms that reflect energy dissipation and maneuver overload constraints. Acceleration magnitude and maneuvering overload are constrained during training to suppress jittery maneuvers. In particular, the normal acceleration required to transition from node $\Delta \boldsymbol{y}_i$ to its neighboring node $\Delta \boldsymbol{y}_j$ is given by

$$a_{\text{req}} = \frac{\boldsymbol{V}_{T,t}}{\Delta t} \arccos \frac{\Delta \boldsymbol{y}_j \cdot \Delta \boldsymbol{y}_i}{\left\| \Delta \boldsymbol{y}_i \right\|}. \tag{45}$$

The energy deficit caused by excessive braking during maneuvering can be calculated as

$$c_r = a_{r,t} - a_{r,\min}. \tag{46}$$

The unitization of acceleration a_{req}, braking energy loss c_r, and jerk j_t yields $\tilde{a}_{\text{req}}$, $\tilde{c}_r$, and $\tilde{j}_t$, respectively. An evaluation mechanism for energy loss and overload limits is defined as follows:

$$s_{ij}^{(E),t} = -\frac{1}{\kappa_E} \log \left(e^{\kappa_E \tilde{a}_{req}^2} + e^{\kappa_E \tilde{j}_t^2} + e^{\kappa_E \tilde{c}^2} \right) \tag{47}$$

where κ_E represents the entropy risk. The energy-related teacher score $s_{ij}^{(\mathrm{E}),t}$ is defined by (47) for the transition from node $\Delta \boldsymbol{y}_i$ to its neighboring node $\Delta \boldsymbol{y}_j$ at time t. The score is constructed by aggregating three normalized physical indicators, namely, the required normal acceleration $\tilde{a}_{\text{req}}$, the braking energy loss $\tilde{c}_r$, and the jerk $\tilde{j}_t$. The tt form provides a smooth and differentiable approximation to the maximum of these indicators, so that the most critical violation among maneuver overload, braking loss, and jerk dominates the aggregated penalty while maintaining stable gradients during training. The scaling factor κ_E is the temperature parameter that controls the sharpness of the LogSumExp aggregation and therefore adjusts the sensitivity of $s_{ij}^{(\mathrm{E}),t}$ to large constraint violations. As a result, neighbor transitions associated with smaller $\tilde{a}_{\text{req}}$, $\tilde{c}_r$, and $\tilde{j}_t$ yield higher scores and are assigned larger attention, which biases the attention-head group $\mathcal{H}_E$ toward energy-saving and smooth maneuver directions.

2) *Pursuit–Evasion decision scoring:* By considering the intercept engagement, the edges associated with penetration risk and miss probability are the primary focus of research. Factors related to the risk of breaking through defense include the distance $\boldsymbol{r}_t$, the escape rate $\dot{\boldsymbol{r}}_t$, and the boundary of the maneuverable reachable domain $\mathcal{P}_{T,t}$. The factors are normalized to obtain the escape risk

$$R_{j,\tau} = \mathrm{LSE}_{\kappa R}\left(\tilde{\boldsymbol{r}}_t, \tilde{\dot{\boldsymbol{r}}}_t, \tilde{\mathcal{P}}_{T,t}\right) \tag{48}$$

where the Log-Sum-Exp mapping $\mathrm{LSE}_\kappa(x_1, \ldots, x_n) = \frac{1}{\kappa} \ln(\sum_{i=1}^n \exp(\kappa x_i))$ is used to smoothly aggregate multiple penalty terms into a single scalar and serves as a differentiable approximation of $\max\{x_1, \ldots, x_n\}$ to emphasize the dominant constraint violation. The teacher evaluation system related to flight risk has been established as

$$s_{ij}^{(R),t} = \frac{1}{\lambda_R} \log \sum_{\tau=t}^{t+H} w_{j,\tau} \exp\left(\lambda_R r_{j,\tau}\right) \tag{49}$$

where $w_{j,\tau}$ denotes the cross-temporal aggregation weight and λ_R represents the risk assessment coefficient, which controls the attention bias toward risky moments. This mechanism can evaluate risks when facing threat under the intercept engagement and assign additional weight to the most dangerous threat. The $\mathcal{H}_R$ feature extraction prioritizes the collection of the state and kinematic information of threat.

3) *Task requirements and timeliness scoring:* The target pursuit–evasion decision under intercept engagement is constrained by mission requirements and time limitations. In practical scenarios, the target is required to follow a specific path corridor within a given time window as much as possible to approach the designated position. The training focused solely on evasion and risk management caused the target to abandon the original mission and deviate from the original trajectory.

The corridor deviation of the target trajectory in both radial and normal directions is calculated based on the central path $\boldsymbol{Y}_k^*$ of the corridor k within the original mission corridor

$$\delta_{r,t}^{(k)} = \left[\mathrm{dist}_r\left(\boldsymbol{y}_t, \boldsymbol{y}_{t,k}^*\right) - b_r\right] / b_r \tag{50}$$

$$\delta_{n,t}^{(k)} = \left[\mathrm{dist}_n\left(\boldsymbol{y}_t, \boldsymbol{y}_{t,k}^*\right) - b_n\right] / b_n \tag{51}$$

where b_r and b_n denote the radial and normal corridor half-widths, respectively. The target deviation from the original mission objective location $\mathcal{Y}$ and time requirement T^* is calculated and normalized, ultimately establishing the teacher scoring

$$\begin{aligned} s_{ij}^{(T),t} = & -\Big[(1-u_t)\, LSE_{\kappa_T}\left(\delta_{r,t}^{(k)}, \delta_{n,t}^{(k)}\right) \\ & + \mu_t LSE_{\kappa_T}\left(\left\|\boldsymbol{y}_t - \mathcal{Y}\right\|, \left|T - T^*\right|\right)\Big] \end{aligned} \tag{52}$$

where T is the total duration of the maneuver trajectory and κ_T is the coefficient. The gating function μ_t is

$$\mu_t = \sigma\left(\kappa_T \frac{t - T^*}{T}\right). \tag{53}$$

In the early stages of interception, a smaller μ_t leads to greater attention being directed toward corridor correction. The increasing μ_t at the time of imminent interception

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gradually shifts focus toward terminal convergence. This aligns with the temporal structure of practical task losses.

REMARK 4 The teacher scores are used only during training to supervise factor-specific attention behaviors and to regularize candidate selection through the interactive cropping module. They are not required as additional inputs during inference, where the predictor relies solely on online observations and engagement states. This design prevents supervision signals from leaking into deployment and ensures that the learned attention mechanism generalizes as a data-driven interaction model.

These evaluation factors vary significantly in their physical dimensions. The uncertainty weighting approach prevents any single factor from dominating others due to excessive influence. The individual teacher scores in (49), (51), and (54) are substituted into (36) and (37) to construct the loss function at the attention layer

$$\mathcal{L}_k^{\text{attn}} = \sum_{k\in\{E,R,T\}} \left(\frac{1}{2\tilde{\sigma}_k^2} \mathcal{L}_k^{\text{attn}} + \log \tilde{\sigma}_k \right) + \lambda_{\text{sep}} \mathcal{L}_{\text{sep}} \tag{54}$$

where $\tilde{\sigma}_k$ is the homoscedasticity scale for the attention group $\mathcal{H}_k$ used to balance each factor. In order to ensure training stability, the subtasks with larger uncertainty and noise are assigned larger σ_k or a smaller weight.

This supervised learning mechanism is integrated with ADE in (34) and FDE in (35) to establish a total loss function for trajectory extrapolation

$$\mathcal{L}_{\text{tra}} = \mathcal{L}_k^{\text{attn}} + \frac{1}{2\bar{\sigma}_{\text{sup}}^2} (\omega_1 \text{ADE} + \omega_2 \text{FDE}) \tag{55}$$

where the σ_{sup} is the covariance measure for ADE and FDE, while ω_1 and ω_2 are the weight coefficients for ADE and FDE, respectively. The attention loss $\mathcal{L}^{\text{attn}}$ affects the gradients of all trainable parameters through backpropagation. The backpropagation paths are shown in Fig. 6, where the red and blue lines represent the geometric path and attention path, respectively. The trajectory extrapolation algorithm based on the distribution cropper and the supervised learning mechanism is shown in Table II. Based on the supervised learning mechanism proposed in this article, the trajectory extrapolation neural network can be trained. In addition, the proposed distribution cropper can influence the trajectory results through the constraint function. The intercept engagement and pursuit–evasion decision are incorporated into the trajectory extrapolation by the proposed algorithm.

The overall trajectory prediction algorithm is summarized in Table III. The proposed method targets realistic terminal interception control under adversarial engagement, where the target and defender adapt their strategies as the engagement evolves. Interception and evasion are modeled as a time-varying closed-loop pursuit–evasion game rather than a unidirectional causality. At each update instant, the latest observations and engagement states are used to extract engagement factors, infer the intention posterior, suppress engagement-inconsistent candidates through interactive cropping and attention propagation on the interaction graph, and generate an extrapolated trajectory consistent with the current engagement state.

TABLE II
GAT With Interactive Crop and Supervised Learning

Algorithm 2

Input: Normalized samples, interaction graph $\mathcal{G}$, and interactive cropper configurations.
Output: Predicted increments $\{\Delta\hat{\boldsymbol{y}}_k\}$ and extrapolated trajectories $\hat{\boldsymbol{Y}}$; updated parameters in training.

Step 1:	Normalize and transform input samples. Construct the interaction graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \varphi_{\text{edg}})$ as formulated in (15).
Step 2:	Perform spatial projection and compatibility scoring. The soft masks and attention weights are computed according to (19)–(21).
Step 3:	Apply interactive distribution cropping under the constraint set $\mathcal{C}$ defined in (22)–(24). The attention redistribution follows (25), and the cropping procedure is implemented by Table I (Algorithm 1).
Step 4:	Propagate messages and update hidden states using the cropped attention weights in (17)–(18).
Step 5:	Decode the future coordinate increments in (32) and integrate them to obtain extrapolated trajectories according to (33).
Step 6:	Evaluate and supervise the extrapolation at both geometric and attention levels. The total loss is given by (55), with ADE and FDE defined in (34) and (35).
Step 7:	Backpropagate gradients along both the geometric and attention paths, which merge at the attention weights α_{ij}.
Step 8:	Update parameters following the training schedule, with gradient clipping $\|\nabla\|_2 \leq \nabla_{\max}$.

REMARK 5 The supervised learning mechanism uses constructed teacher scores as factor-specific supervisory signals to guide attention learning under physical and mission constraints. These scores are derived from engagement-related information to provide differentiable targets associated with energy dissipation, interception risk, and task timeliness. Attention assigns data-dependent importance to interaction information, and disjoint head grouping enables factor-specific supervision without gradient entanglement. Divergence-based regularization aligns the learned attention distribution with the teacher-induced target distribution in a smooth and stable manner, while entropy-related terms prevent attention collapse and promote calibrated attention patterns. Uncertainty weighting balances heterogeneous loss components by learning their relative weights, reducing manual tuning, and preventing any single term from dominating training.

The algorithm runs online with a short update interval, periodically refreshing the observation window and engagement states to produce a continuously updated sequence of forecasts. Each iteration re-estimates the intention posterior and reapplies interactive cropping and extrapolation, so transient recognition deviations are corrected as new measurements arrive rather than being propagated as fixed

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TABLE III
Online Trajectory Prediction With Intention Recognition and Intercept Engagement

Algorithm 3

Input: Streaming observations $\boldsymbol{X}$, defender state $\boldsymbol{s}_M$, engagement state $\boldsymbol{s}_L$, and trained networks for intention recognition and trajectory extrapolation.
Output: Trajectory prediction is performed iteratively at time intervals of $\hat{\boldsymbol{Y}}(t_k)$ over a horizon of length t_p at each update instant t_k.

Step 1:	**Initialization.** Set the prediction update interval ΔT_p and initialize the observation window ending at t_0.
Step 2:	**Prediction update.** At each update instant t_k with spacing ΔT_p, refresh the observation window $\boldsymbol{X}(t_k)$ and the associated states $\boldsymbol{s}_M(t_k)$ and $\boldsymbol{s}_L(t_k)$.
Step 3:	**Intention recognition.** Compute the intention posterior according to (12)–(14), where the defender state encoding is incorporated by (13).
Step 4:	**Interactive graph construction and fusion.** Construct the interaction graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \varphi_{\text{edg}})$ as formulated in (15), and fuse node features using (16).
Step 5:	**Interactive distribution cropping.** Apply the constraint set $\mathcal{C}$ defined in (22)–(24) and redistribute attention masks following (25); the cropping procedure is implemented by Table I (Algorithm 1).
Step 6:	**Trajectory extrapolation.** Compute attention weights according to (19)–(21) and update hidden states using (17)–(18).
Step 7:	**Decoding and integration.** Generate the future coordinate increments in (32) and integrate them to obtain the extrapolated trajectory according to (33), yielding $\hat{\boldsymbol{Y}}(t_k)$ for $k = 1, \ldots, t_p$.
Step 8:	**Training note.** During offline training, the trajectory extrapolation is supervised by the total loss given by (55), with ADE and FDE defined in (34) and (35); during online inference, only the forward computation in Steps 3–7 is executed.
Step 9:	**Iteration.** Set $t_{k+1} \leftarrow t_k + \Delta T_p$ and repeat Steps 2–8 until termination.

inputs. The short update interval further keeps the forecasts aligned with rapidly evolving engagement geometry, ensuring real-time responsiveness and control relevance even with multistep prediction horizons.

IV. NUMERICAL SIMULATION RESULTS

A. Maneuver Intention Recognition Based on BiLSTM

The proposed intention recognition algorithm processes 6-D kinematic time series composed of position and velocity components. The input embedding dimension is set to $d = 6$. A bidirectional LSTM is adopted as the recurrent backbone with hidden size $N_h = 128$, and the hidden state at the final time step is used to summarize the window-level dynamics. To stabilize hidden-state statistics and improve optimization, layer normalization is applied after the recurrent block. Dropout with probability $p = 0.2$ is employed both before and after the intermediate projection. The intermediate fully connected layer has width $m = 64$ with a ReLU activation. The classification head maps the latent representation to nine maneuver categories and outputs class probabilities through a Softmax layer. Cross-entropy loss is used for supervised training.

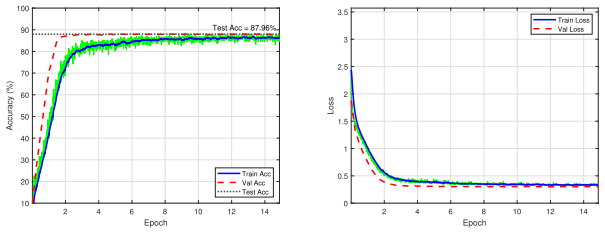
To evaluate the intention recognition algorithm, a 3-D target–defender engagement scenario is simulated with sampling interval $\Delta t = 0.5$ s. Sliding windows of length $T_o = 10$ s are extracted with stride $s = 5$ to form supervised sequences, resulting in approximately 5×10^4 samples. This construction provides broad coverage of LOS orientations and lateral–vertical coupling patterns. A track-level split of 70%, 15%, and 15% is used for training, validation, and testing, respectively. Standardization parameters μ and σ are computed on the training partition and then fixed for validation and testing to ensure a consistent evaluation protocol.

The network is optimized using Adam with an initial learning rate $\eta_0 = 10^{-3}$. A stepwise learning-rate schedule is applied with decay factor $\gamma = 0.5$ at epoch 30. The batch size is set to 256 and the model is trained for 70 epochs. Gradient clipping with threshold 1.0 is used to mitigate exploding gradients and improve training stability. Training samples are shuffled at every epoch, and validation performance is recorded at the end of each epoch. For visualization, per-iteration logs are aligned to the epoch axis, and an exponential moving-average curve is overlaid to improve readability. The test accuracy is additionally indicated by a horizontal reference line.

Supervision is derived from an intercept engagement and evasion-maneuver mechanism. The maneuver command is constructed along the anti-LOS direction and projected onto the velocity-normal plane to ensure physical realizability. Target maneuvers follow naval air training and operating procedures standardization (NATOPS)-based primitives and commonly used maneuvering patterns reported in [42], [44], [45], [46], and [47]. The detailed dataset construction, including operator forms and activation logic, is provided in Section IV-B. Each trajectory is synthesized by concatenating short maneuver primitives, including coordinated turns, climb, dive, and their coupled composites, under kinematics and overload saturation. Window-level labels are assigned by the dominant primitive semantics to form the discrete intention space for BiLSTM recognition.

The intention space for maneuver recognition is defined as a finite semantic set comprising nine classes $\boldsymbol{I} = \{$*no maneuver*, *left turn*, *right turn*, *climb*, *dive*, *left-climb*, *right-climb*, *left-dive*, *right-dive*$\}$. Labels are assigned at the window level to represent the dominant maneuver semantics within each observation window. Each trajectory is generated under a pursuit–evasion mechanism, where terminal evasion is triggered once the separation distance decreases to a prescribed fraction of the initial distance. The detailed maneuver generation mechanism is presented in Section IV-B. Maneuver primitives are concatenated to construct coordinated turn, climb, and dive segments as well as their composite patterns, while both segment duration and overload magnitude are sampled from bounded ranges to cover mild and aggressive behaviors without introducing unrealistic outliers. The engagement geometry is diversified by varying the initial separation, relative azimuth, relative elevation, and relative heading. The defender aircraft is initialized both inside and outside the target's FOV to ensure

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(a) (b)

Fig. 8. Maneuver intention space training. (a) Training accuracy. (b) Training loss.

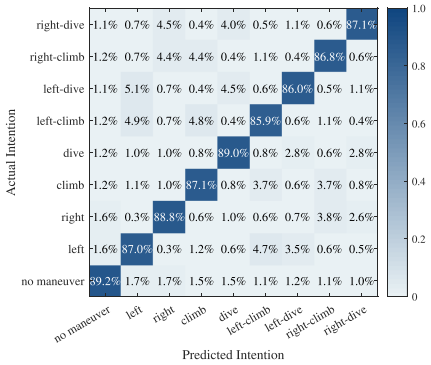
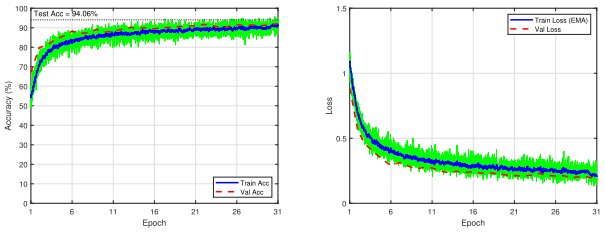


Fig. 9. Maneuver intention recognition test confusion matrix.



(a) (b)

Fig. 10. Acceleration intention space training. (a) Training accuracy. (b) Training loss.

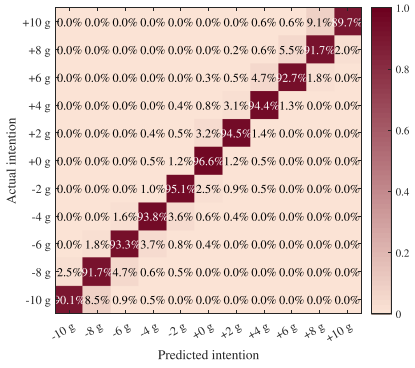


Fig. 11. Acceleration intention recognition test confusion matrix.

broad coverage over LOS orientations and lateral–vertical coupling patterns.

The confusion matrices are computed on the held-out test partition under a unified evaluation protocol. For each maneuver label, 1000 independent Monte Carlo trials with distinct random seeds are conducted to form the final test set. Training outcomes are reported in Figs. 8 and 9. Under the unified protocol, the intention recognition algorithm exhibits a rapid increase in accuracy during early training, followed by gradual convergence. Both training and validation cross-entropy decrease monotonically with closely aligned trajectories, indicating stable optimization under the track-level split and train-only standardization. The test accuracy reaches approximately 88%, and the resulting confusion matrix is strongly diagonally dominant, with compact off-diagonal mass primarily concentrated among semantically adjacent classes. Minor cross-confusions are observed between no maneuver and mild turns, as well as between left or right maneuvers and their climb or dive counterparts, whereas opposite-direction pairs and vertical opposites, such as climb versus dive, exhibit negligible confusion. Overall, the matrix structure suggests that the engagement-aware formulation captures salient spatiotemporal cues of maneuver semantics, and that the BiLSTM-based intention recognition algorithm provides reliable discrimination across all nine classes, thereby serving as a robust front-end for subsequent online iterative prediction and intercept-decision modules.

Beyond the discretized maneuver semantics, the proposed intention recognition algorithm also supports an acceleration-based formulation, in which maneuver intensity is quantized into ordered normal-acceleration levels to define an alternative intention space. Specifically, the normal acceleration is discretized into 11 ordered levels, $\mathcal{I} = \{-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10\}$ g, where $\mathrm{g} = 9.81\ \mathrm{m/s^2}$. The recognition statistic is defined as the mean overload $\bar{a}_n$ within an observation window, computed by $\bar{a}_n = \frac{1}{t_o}\sum_{k=1}^{t_o} a_n(k)$. The window label is then assigned by locating $\bar{a}_n$ in the interval $(I_m - 0.5,\ I_m + 0.5]$ for some $I_m \in \mathcal{I}$, where boundary ties are consistently mapped to the larger label.

Training results are reported in Figs. 10 and 11. Under the unified protocol, the classification accuracy increases rapidly at the early stage and then transitions to a smooth saturation around the midtraining epochs. Meanwhile, the training and validation cross-entropy decrease monotonically with closely matched trajectories, which indicates a stable optimization process. On the held-out test set, the intention recognition algorithm achieves approximately 94% accuracy. The row-normalized confusion matrix is dominated by a pronounced diagonal band, accompanied by a narrow neighboring band that is consistent with the ordered nature of the labels. Midlevel bins typically attain 94%–96%, whereas extreme bins exhibit mild inward spill, which aligns with finite support and saturation effects. Overall, the matrix structure corroborates that the engagement-aware formulation, coupled with BiLSTM temporal encoding, resolves maneuver intensity with fine granularity across all 11 levels while preserving coherent training and validation dynamics.

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TABLE IV
Configurable Parameters and Ranges for Scenario Sampling in the Maneuver-Intention Sample Library

Configurable parameter	Range
Initial separation distance	$r_0 \in [4,\ 8]\,\mathrm{km}$
Initial LOS yaw angle	$q_{\varphi,0} \in [-180^\circ,\ 180^\circ]$
Initial LOS pitch angle	$q_{\vartheta,0} \in [-30^\circ,\ 30^\circ]$
Initial altitude difference	$\Delta h_0 \in [-1,\ 1]\,\mathrm{km}$
Initial target speed	$V_T \in [250,\ 350]\,\mathrm{m/s}$
Initial defender speed	$V_M \in [380,\ 450]\,\mathrm{m/s}$
Yaw maneuver overload magnitude	$a_{\varphi,0} \in [-88.3,\ 88.3]\,\mathrm{m/s^2}$
Pitch maneuver overload magnitude	$a_{\vartheta,0} \in [-88.3,\ 88.3]\,\mathrm{m/s^2}$

B. Trajectory Extrapolation Comparisons and Ablation Studies

Trajectory extrapolation is performed based on the available observed trajectory together with the intercept engagement context. Training samples are generated from a typical maneuver dataset. To emulate representative noncooperative terminal evasive behaviors, once the evasion condition is triggered, the target is commanded to produce an overload opposite to the LOS direction.

Table IV summarizes the configurable parameters used to instantiate diversified 3-D engagement scenarios and to parameterize the engagement-driven maneuver command. The initial separation distance r_0 and the initial LOS yaw and pitch angles $q_{\varphi,0}$ and $q_{\vartheta,0}$ jointly determine the initial relative geometry, while the initial altitude difference Δh_0 specifies the vertical offset between the target and the defender. The initial target and defender speeds V_T and V_M set the kinematic scale of the engagement. The maneuver overload magnitudes $a_{\varphi,0}$ and $a_{\vartheta,0}$ specify the yaw-plane and pitch-plane maneuvering authority used to generate the evasive response after the evasion trigger, subjected to the target overload limit. Based on these sampled parameters, the engagement scenario is propagated and the evasion decision mechanism is applied to produce physically realizable trajectories and window-level intention labels.

The evasion command is constructed by first prescribing an anti-LOS acceleration direction $-\boldsymbol{e}_L(t)$ and then projecting it onto the plane normal to the target velocity to obtain a realizable overload command. The commanded evasive acceleration is computed via overload saturation

$$\tilde{\boldsymbol{a}}_{\mathrm{eva}}(t) = -\boldsymbol{e}_L(t)\left[\boldsymbol{I} - \boldsymbol{e}_T(t)\boldsymbol{e}_T^{\mathrm{T}}(t)\right]$$
$$\boldsymbol{a}_{\mathrm{eva}}(t) = a_{\mathrm{eva}}\frac{\tilde{\boldsymbol{a}}_{\mathrm{eva}}(t)}{\max\{\|\tilde{\boldsymbol{a}}_{\mathrm{eva}}(t)\|_2,\ \epsilon\}}$$

where $a_{\mathrm{eva}} = \min\{\bar{a}_{\mathrm{eva}},\ a_{T,\max}\}$, $\bar{a}_{\mathrm{eva}}$ is the nominal evasion overload level, and $\epsilon = 10^{-6}$ is a small constant to avoid numerical singularity. This anti-LOS overload mechanism reflects a typical terminal evasion strategy in realistic adversarial engagements. The target biases its maneuver normal to the velocity direction while steering away from the defender aircraft along the LOS geometry, subjected to physical overload limits. The evasion maneuver is generated by holding $\boldsymbol{a}_{\mathrm{eva}}(t)$ within each short primitive segment and updating it at segment transitions, yielding a kinematically continuous trajectory under the same discrete-time propagation. Segment duration and the nominal overload level $\bar{a}_{\mathrm{eva}}$ are sampled within predefined bounds, with feasibility enforced by saturation at $a_{T,\max}$. Since $\boldsymbol{a}_{\mathrm{eva}}(t)$ is determined by $\boldsymbol{e}_L(t)$ and $\boldsymbol{e}_T(t)$, lateral and vertical components are induced by engagement geometry rather than random concatenation.

The target activates engagement-driven evasion when $r \le \frac{2}{3}r_0$, where r_0 is the initial distance between the target and the defender at time T_p. This trigger rule normalizes the maneuver-onset timing across scenarios with different initial separations by using a fixed fraction of the initial distance. It places evasion in the terminal portion of the engagement while still leaving enough time for the proposed online loop to update the intention posterior and reapply engagement-consistent cropping after maneuver onset. Setting the trigger much earlier would over-emphasize long-range segments with weak interaction cues, whereas setting it much later would leave too few updates to evaluate engagement-conditioned adaptation. The database is explicitly constructed to cover engagement geometries with the defender aircraft both inside and outside the target's FOV. Therefore, when the defender aircraft is outside the visible sector and is unobservable, the predictor does not introduce spurious interaction-driven responses and instead naturally degenerates to an extrapolation behavior consistent with the ablation setting without intercept engagement coupling.

A maneuvering database is constructed by following NATOPS-style maneuver patterns and a pursuit–evasion decision mechanism based on trajectories generated by the evasive acceleration. The trajectory extrapolation dataset comprises 50 000 instances and is partitioned into an 80% training set and a 20% validation set. Each instance corresponds to a complete engagement trajectory segment consisting of synchronized time series for the target and defender kinematic states sampled at a fixed interval, together with the associated engagement descriptors used by the proposed model. Each trajectory is generated by concatenating maneuver primitives under the same kinematics and overload saturation, covering coordinated turns, climb, dive, and their coupled composites. Maneuver classes are sampled with balanced coverage, while overload magnitude and maneuver duration are sampled from bounded distributions subjected to the target overload limit, representing both mild and aggressive behaviors without unrealistic outliers. Engagement geometry is further diversified by sampling the initial separation, aspect angle, elevation difference, and relative heading in three dimensions, with the defender aircraft initialized inside the target FOV to ensure broad coverage of LOS orientations and engagement-induced lateral–vertical coupling patterns.

Table V summarizes the end-to-end procedure for constructing the maneuver-intention sample library. The process for generating the dataset used for training is illustrated in Fig. 12. Scenario parameters are sampled within the specified ranges, a 3-D terminal engagement is instantiated and propagated under the discrete-time kinematic model, and the engagement-driven evasion command is applied

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TABLE V
Sample Dataset Generation Process

Stage	Procedure
A1	Sample initial conditions and maneuver parameters within predefined ranges to ensure diversified engagement geometry.
A2	Instantiate a terminal engagement scenario using the sampled parameters and initialize target and defender states.
A3	Propagate both agents using the discrete-time point-mass kinematics and update LOS-related engagement states.
A4	Apply the evasion decision mechanism. Generate target evasive maneuvers $\boldsymbol{a}_{\mathrm{eva}}(t)$ under overload limits.
A5	Extract sliding windows from the generated trajectories and assign window-level maneuver-intention labels in the nine-class semantic intention space.
A6	Append labeled windows to the intention-sample library. Repeat A1–A6 until the target dataset size and coverage are reached.

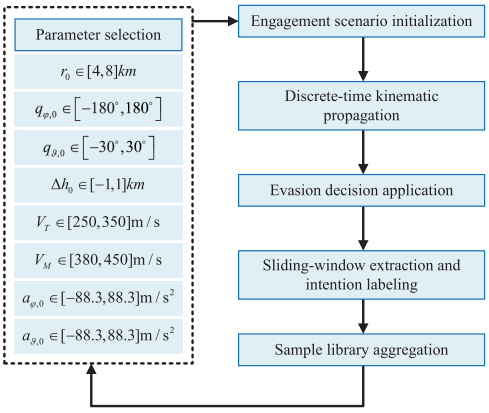
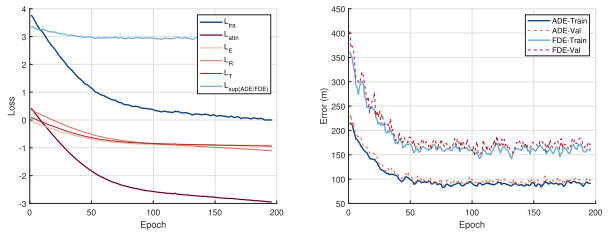


Fig. 12. Dataset generation process.

once the trigger condition is satisfied. Sliding windows are then extracted from each trajectory, and each window yields a supervised sample consisting of an observed kinematic sequence and a nine-class maneuver-intention label. The sampling–simulation–labeling loop is repeated until the target dataset size and geometry coverage are achieved.

The connection strategy for the target and defender aircraft interactive graph is based on radius. The Top-K cropping is applied to each source node with $K = 8$. The edge feature is established by φ_{edg} shown in (15) and the path constraints imposed by the intercept engagement. The interactive graph dynamically reconstructs itself and imposes hard blocking on links deemed unreachable. The intensity of soft cropping and hard cropping increases with each training iteration during the interactive distribution cropping.

Configure the parameters for the GAT. In the spatial layer, the embedding dimension $d = 256$, the number of attention heads $H = 6$, and both the key vector dimension d_k and the value vector dimension d_v are 64. At the temporal layer, the number of attention heads $H = 4$ and the dimensions of key and value vectors are identical to those of the spatial layer. The attention structure employs a layer depth of 2 for both temporal and spatial dimensions. The FOV mask is enabled during interaction and the attention temperature $\tau = 0.1$. The decoder employs the autoregressive GRU and executes cross-attention over temporal memory. The linear regression is employed for the output head. The parameters in the loss function (55) are $\lambda_{\mathrm{sep}} = 0.1$, $\omega_1 = 0.4$, and $\omega_2 = 0.6$. The AdamW optimizer is selected



(a) (b)

Fig. 13. Training process of trajectory prediction. (a) Training loss. (b) Training error.

and uncertainty weighting is implemented using Kendall heteroscedasticity. The heteroskedasticity parameter range is constrained to $[-6, 6]$ to prevent training results from diverging. AdamW is used with initial learning rate 1×10^{-3}, weight decay 1×10^{-4}, and gradient clipping at 1.0 to stabilize training. The batch size is set to 256. The model is trained for 200 epochs. A cosine decay schedule is applied after a short warm-up of 5 epochs. Early stopping monitors the validation metric ADE and FDE. Training is terminated if the validation ADE and FDE does not improve by at least 1 m for 20 consecutive epochs, and the best-performing checkpoint on validation is selected for reporting.

Input information includes the existing trajectory and the kinematic state of the defender aircraft at time $T_o = 8$ s. The training process is shown in Fig. 13. The model rapidly learns reasonable interactive attention patterns under multifactor constraints. The proposed engagement-aware supervision consistently reduces each loss component during training, indicating more stable optimization across factor-specific objectives. The research findings can effectively counter noncooperative maneuverable target. The ADE and FDE converge from hundreds of meters to approximately 10^2 m level. The training results show that the proposed method converges with a lower total loss and maintains balanced optimization across multitask sublosses via heteroscedastic weighting. The trajectory extrapolation maintains acceptable terminal accuracy within the prediction window. Simulation results validate the effectiveness and robustness of the designed GAT and multitask loss in multifactor interaction environment.

A 3-D terminal engagement scenario is considered in an inertial frame. Both the target and the defender aircraft are propagated by a discrete-time point-mass kinematic update with sampling interval $\Delta t = 0.1$ s. The observation window is set to $T_o = 8$ s, and the prediction horizon is set to $T_p = 6$ s. The simulation presents the output of a single iteration within the online iterative prediction process, in order to illustrate the extrapolation behavior under a given observation window and engagement state. For clarity of presentation, the prediction horizon is set to 6 s and the corresponding forecast trajectory is plotted. In practical interception applications, however, the predictor is updated online at a short update interval, and the refresh rate is higher than the horizon length. Therefore, the update interval between two consecutive predictions is shorter than

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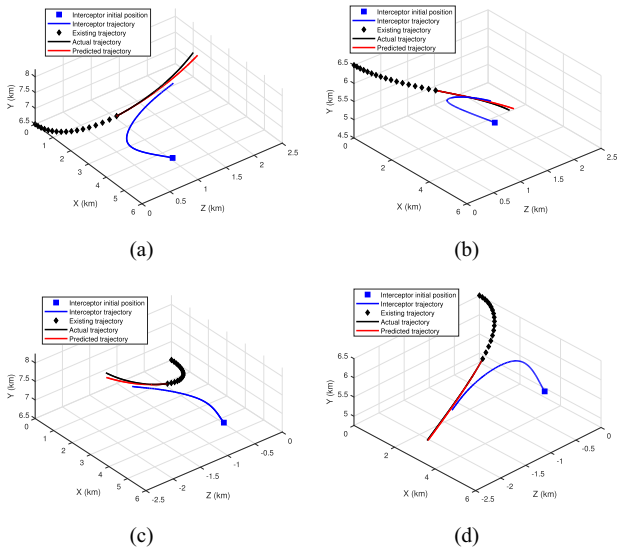


Fig. 14. Trajectory prediction based on intercept engagement. (a) Left-climb. (b) Left-dive. (c) Right-climb. (d) Right-dive.

TABLE VI
Initial Conditions of Four Composite Maneuver Scenarios

Scenario	$a_{\vartheta,0}$ (m/s^2)	$a_{\varphi,0}$ (m/s^2)	r_0 (km)	$\boldsymbol{e}_{L,0}$
Left–Climb	39	−58	5.3	$[0.98, -0.15, 0.09]^{\mathrm{T}}$
Left–Dive	−39	−58	5.3	$[0.98, -0.15, -0.09]^{\mathrm{T}}$
Right–Climb	39	58	5.3	$[0.98, 0.15, 0.09]^{\mathrm{T}}$
Right–Dive	−39	58	5.3	$[0.98, 0.15, -0.09]^{\mathrm{T}}$

6 s, allowing the model to continuously incorporate new measurements and to revise the forecast in response to the rapidly evolving engagement.

Target and defender aircraft speeds are set to $V_T = 300\,\mathrm{m/s}$ and $V_M = 360\,\mathrm{m/s}$, respectively. The target maximum overload is limited by $a_{T,\max} = 9\,\mathrm{g}$, and the defender aircraft overload limit is set to $a_{M,\max} = 15\,\mathrm{g}$. A FOV angle constraint is imposed on the target with half-angle $\theta_{\mathrm{FOV}} = 60^\circ$, and the defender aircraft is initialized to approach from within the target visible sector throughout the engagement.

Let the actual positions be $\{\boldsymbol{y}_t\}_{t=1}^{t_p}$ and the predicted positions be $\{\hat{\boldsymbol{y}}_t\}_{t=1}^{t_p}$, where $t_p = T_p/\Delta t$. The prediction error is defined as

$$e(t) = \left\|\hat{\boldsymbol{y}}_t - \boldsymbol{y}_t\right\|_2, \quad t = 1, \ldots, t_p.$$

After activation, the evasion direction is determined by the LOS geometry. A maneuvering database for adversarial games is established based on actual combat requirements. Maneuver selections are made according to typical maneuver patterns observed in real-world scenarios. The resulting acceleration is finally saturated by $a_{T,\max}$ to maintain feasibility without introducing actuator lag. Table VI summarizes the four maneuver scenarios and their corresponding initial configurations. The interceptor is modeled as the defender aircraft in the engagement. Fig. 14 illustrates the trajectory extrapolation results under four composite maneuvers.

Fig. 14 reports the proposed predictor under four composite maneuvers. The terminal errors corresponding to trajectory predictions in each scenario are 150 , 202 , 81 , and 173 m. The predicted trajectories consistently follow the actual trajectory, including both the lateral turning tendency and the coupled climb and dive component. Across the four cases, the predicted trajectory remains close to the ground truth throughout the horizon. The proposed method generalizes across different maneuver quadrants without requiring case-specific tuning. This consistency provides the basis for the subsequent comparative evaluation.

Representative sequence-modeling baselines are considered, including an LSTM predictor following [34], a GRU predictor following [40], a temporal convolutional network following [48], and a Transformer encoder–decoder following [12]. The LSTM baseline models temporal dependencies through gated memory updates to retain long-term motion context, while the GRU baseline adopts a simplified gating structure to achieve efficient recurrent modeling with fewer parameters. The temporal convolutional network (TCN) baseline performs causal temporal feature extraction using stacked 1-D convolutions with a large effective receptive field, thereby capturing multiscale history without recurrence. The Transformer baseline employs self-attention to model global dependencies across the observed sequence and decodes future states through attention-based sequence-to-sequence mapping. These baselines extrapolate future trajectories primarily from the observed motion history and do not explicitly incorporate intention recognition or intercept engagement, which makes them prone to accumulated lateral and vertical deviation once terminal evasion becomes dominant.

For the neural-network baselines, LSTM, GRU, TCN, and Transformer are implemented under a matched-capacity setting to ensure a fair comparison. Specifically, both LSTM and GRU adopt a 2-layer encoder–decoder architecture with hidden width $n_{\mathrm{hid}} = 256$ and dropout rate $p_{\mathrm{do}} = 0.1$. The TCN baseline employs four residual temporal-convolution blocks with kernel size $k_{\mathrm{ker}} = 3$, channel width $c_{\mathrm{ch}} = 256$, and dilation set $\mathcal{D} = \{1, 2, 4, 8\}$, followed by a linear projection head; the dropout rate is set to $p_{\mathrm{do}} = 0.1$. The Transformer baseline is implemented as a 2-layer encoder–decoder with model width $d_{\mathrm{mdl}} = 256$, $h_{\mathrm{att}} = 4$ attention heads, feedforward width $d_{\mathrm{ff}} = 1024$, and dropout rate $p_{\mathrm{do}} = 0.1$. All baselines output future positions over the same prediction horizon as the proposed method.

REMARK 6 The interacting multiple model (IMM) filter is suitable for trajectory prediction involving multiple maneuver transitions in certain scenarios. However, the IMM-based algorithm demonstrates significant limitations for extrapolation beyond a 2–3 s. The accuracy declines rapidly after this short time frame, resulting in substantial prediction errors. Therefore, due to its inadequate performance for longer-term predictions, IMM was not included as a baseline method in the comparative analysis.

Figs. 15 and 16 compare the proposed method with LSTM, GRU, TCN, and Transformer baselines. First, all

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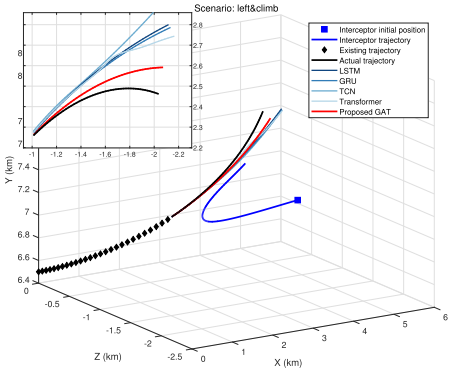
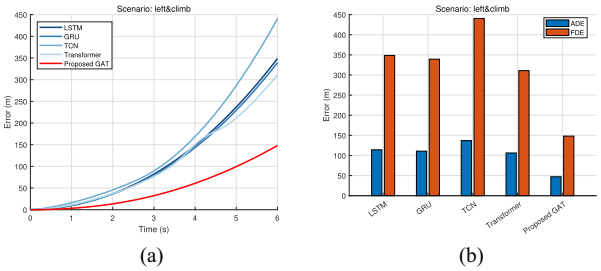


Fig. 15. Comparative simulations.



(a) (b)

Fig. 16. Errors in comparative simulations. (a) Prediction error. (b) ADE and FDE.

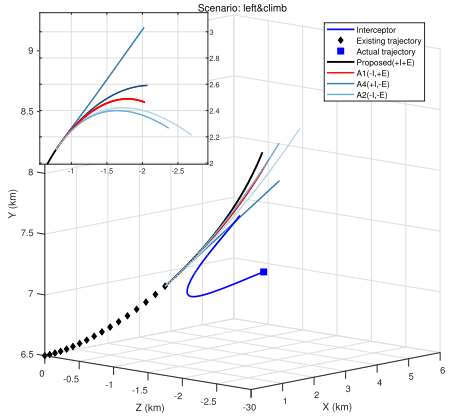
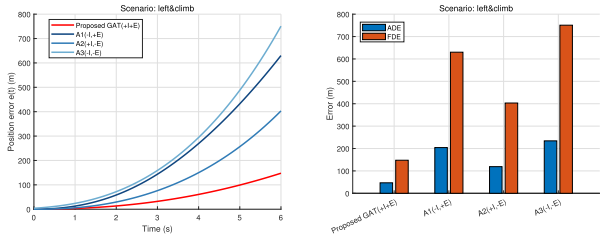


Fig. 17. Ablation simulations.



(a) (b)

Fig. 18. Errors in ablation simulations. (a) Prediction error. (b) ADE and FDE.

trajectory-driven baselines exhibit a faster error accumulation along the horizon. Although some baselines remain competitive in the first portion of the horizon, their $e(t)$ curves grow more rapidly and lead to larger terminal deviations, as reflected by higher FDE. This indicates that purely trajectory-based extrapolation becomes increasingly unreliable when the target motion enters a highly nonstationary regime in the terminal phase. Second, the proposed method yields a consistently smaller error curve and the lowest ADE and FDE. Notably, the advantage is not restricted to a single instant but persists across the entire horizon, which implies that the proposed predictor improves both in ADE and FDE. The lower slope of $e(t)$ further suggests improved long-horizon stability rather than only short-horizon fitting. Geometrically, the baseline methods exhibit lateral drift and gradually deviate from the engagement-consistent curvature, whereas the proposed method preserves the correct deflection tendency and prevents systematic lateral bias.

To further quantify the contribution of each design component, an ablation study is conducted by systematically enabling or disabling intention conditioning (I) and engagement coupling (E). The complete model, denoted as *Proposed GAT* $(+I, +E)$, activates both intention recognition and intercept engagement. Variant A1 $(-I, +E)$ disables intention recognition while retaining engagement coupling. Variant A4 $(+I, -E)$ retains intention recognition but removes engagement coupling. Variant A2 $(-I, -E)$ disables both components and thus reduces to a trajectory-only extrapolation based on GAT.

Figs. 17 and 18 report the corresponding $e(t)$ present the ablation results, comparing the full model (Proposed GAT, $+I, +E$) with A1 $(-I, +E)$, A4 $(+I, -E)$, and A2 $(-I, -E)$. The full model achieves the smallest ADE and FDE in the tested scenario, and the performance degrades monotonically as information components are removed. When engagement coupling is removed (A4, $+I, -E$), the prediction tends to extend preterminal motion trends and becomes less consistent with the terminal maneuver geometry. This results in a noticeable increase in FDE, indicating that the terminal mismatch is dominated by accumulated bias rather than isolated fluctuations. When intention conditioning is removed (A1, $-I, +E$), the model retains engagement-related selection but lacks maneuver-mode priors, leading to increased pointwise deviations over the horizon and thus a higher ADE. Removing both components (A2, $-I, -E$) yields the largest errors, approaching the behavior of a trajectory-only extrapolation backbone. These results confirm that intention conditioning and engagement coupling contribute complementary improvements. The intention recognition stabilizes maneuver representation and reduces in-horizon dispersion, and the intercept engagement mitigates systematic drift in terminal confrontation and improves terminal accuracy.

V. CONCLUSION

This article presents a decoupled trajectory-prediction framework for maneuverable noncooperative targets by integrating an interactive distribution cropper with engagement-aware supervision. First, the trajectory prediction was decoupled and composed of intention recognition and trajectory extrapolation. The independent intention

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recognition was implemented by BiLSTM, considering intercept engagement and pursuit–evasion decision. Second, the interactive distribution cropper was applied in the GAT for target trajectory extrapolation to crop out unreasonable results. Third, a supervised learning mechanism for the trajectory prediction training was proposed based on target–defender dynamics to enhance prediction accuracy. Finally, the proposed trajectory prediction algorithm was validated through numerical simulations to demonstrate its effectiveness and feasibility against maneuverable noncooperative targets.

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